

TCS IPA

Java Question Bank

112 Questions
For study and print use

Question 1

Reverse of a string
Arijit = tijirA

Question 2

Check a string is palindrome or not
Input: pop
Output: Palindrome
Input: Papa
Output: None

Question 3

Write a Java program to check if a given number is an Armstrong number or not.
Here's an example of the expected input and output:
Input number: 153
Output: "Yes, the number is an Armstrong number."

Question 4

Check a number is perfect or not
Input : 6
Output : Perfect Number
Explanation

The divisors of 6 = 1,2,3. $1+2+3 = 6$
Sum of the factors is equal to the given number. So this is perfect
Input : 8
Output : Not Perfect Number
Explanation

Factors of 8 = 1,2,4, but $1+2+4 = 7$ not equal to 8.

Question 5

Write a Java program that takes a user input integer n and prints the Fibonacci series up to n.
The Fibonacci series is a series of numbers in which each number is the sum of the two preceding numbers.
The first two numbers in the series are 0 and 1. For example, the first 10 numbers in the Fibonacci series are:
0, 1, 1, 2, 3, 5, 8, 13, 21, 34.
The program should use a for loop to generate the series.
Sample Input:
20
Sample Output:
0 1 1 2 3 5 8 13

Explanation:

The Fibonacci series up to 20 is: 0, 1, 1, 2, 3, 5, 8, 13. Therefore, the output is "0 1 1 2 3 5 8 13".

Question 6

Write a Java program that takes two user input integers m and n and prints the multiplication table from 1 to n for the number m. The program should use nested for loops to generate the table.

Sample Input:

3 5

Sample Output:

3 x 1 = 3

3 x 2 = 6

3 x 3 = 9

3 x 4 = 12

3 x 5 = 15

Explanation:

The input integers are m=3 and n=5. The program generates the multiplication table for 3 from 1 to 5.

Question 7

Write a Java program that takes a user input integer n and calculates the sum of the first n natural numbers.

The program should use a for loop to calculate the sum.

Sample Input:

5

Sample Output:

The sum of first 5 natural numbers is 15.

Explanation:

The first 5 natural numbers are: 1, 2, 3, 4, 5

Their sum is: 1 + 2 + 3 + 4 + 5 = 15

Question 8

Count number of even digit in a number

If the count greater than or equal to 3, then print "True" otherwise "False"

Question 9

Take a number. If it odd then reverse it, otherwise print "Can not reverse"

Input : 123

Output : 321

Input : 234
Output : Can not reverse

Question 10

Write a Java program to find the sum of all even numbers in an array of integers.

Here's an example of the expected input and output:

Input array: {10, 5, 20, 8, 13}
Output:
Sum of even numbers: 38

Question 11

Write a function that takes a list of numbers as input and returns the sum of all the even numbers in the list.

For example:

Enter size of Array: 4
Create an array:
10 15 20 25
The array is: [10, 15, 20, 25]
Sum of even numbers in the array: 30

Question 12

Count number of vowel, consonent and digit into a string

Question 13

Find number of lower case in a string
AriJIIt = 3

Question 14

Find number of upper case in a string
AriJIIt = 3

Question 15

Count number of word in a string
Note : Always ignore the spaces if more than 1 space given between two word
Sample Input : 1

Kalyani Goverment Engineering College
Process

[Kalyani, Goverment, Engineering, College]
Output

Number of word : 4

Sample Input : 2

Kalyani Goverment Engineering College

Process

[Kalyani, Goverment, , Engineering, College]

Output

Number of word : 4

Question 16

Count number of spaces and characters in a string

Input

Arijit is bad boy

Output

Number of space : 3

Number of character : 14

Question 17

Find the Maximum Ascii valued character in a String

Arijit = t

priyanka = y

Aa = a

Question 18

Minimum Ascii valued character in a String

Question 19

Remove Vowel From A String

Input : Arijit Ghosh

Output : rjt Ghsh

Question 20

Remove a character from a string

Input

Arijit

i

Output

Arjt

Question 21

Find first letter of each word in a string

Input : Kalyani Govt. Eng. College

Output : KGEC

Question 22

Last Character of each word of a sentence

Input: Kalyani Government Engineering College

Output: itge

Question 23

Given a sentence, Find words that start with a capital letter.

For example, in the sentence "The quick Brown Fox jumps over the lazy dog", there are three words that start with a capital letter: "The", "Brown", and "Fox".

Question 24

Find largest word from a given sentence

Input: He is so funny boy

Output: funny

Question 25

Sort name by dictionary order

Input

Arijit

Minaz

Arnab

Output

Arijit

Arnab

Minaz

Question 26

Find Characters of a string at odd index

Arijit = rjt

priyanka = ryna

Question 27

Find the characters in odd position of a string which are consonent

Input: Arijit Ghosh

Output: hs

Question 28

Find occurrence of a character in a string

Input

Priyanka

a

Output

2

Question 29

Find the occurrence of each character in a string

Input: Hello world

Output:

H: 1

e: 1

l: 3

o: 2

w: 1

r: 1

d: 1

Question 30

Find the index number of a character in a String

Input1 :

Carrot

a

Output:

1

Input2:

Carrot

z

Output:

NA

Question 31

Remove duplicate characters in a string

Input : Arijit Ghosh

Output : Arijt Ghos

Question 32

You are given a string *S* consisting of lowercase letters. Your task is to remove all the consecutive duplicates from the string and output the result. For example, if the input string is "abbcddeff", the output should be "abcdef".

Input: abbcdddeff

Output: abcdef

Question 33

Count the words in a sentence which are start with vowel.

Input: Arijit is student of MCA

Output: 3

Input: Sara plays tennis

Output: No such words

Question 34

Find the palindrome words from the sentence and print them and also count

Input:

My name is nitin and I can speak malayalam

Output:

nitin

I

malayalam

3

Question 35

Count the number of words in a sentence that start and end with the same letter.

Input: Anna asked about the Ginseng recipe

Output: 2

Explanation: There are two words in the sentence that start and end with the same letter - "Anna" and "Ginseng".

Question 36

Count the number of words in a sentence that contain at least two consecutive vowels (a, e, i, o, u) in them.

Here's a sample input and output:

Input: I enjoy eating spaghetti and meatballs for dinner

Output: 2

Explanation: There are two words in the sentence that contain at least two consecutive vowels - "enjoy" and "meatballs".

Question 37

Merge two strings using comma

Input:

Arijit,Ghosh

Pop:Mom

Output:

Arijit, Ghosh, Pop:Mom

Question 38

Input : Home, Sweet Home

Home

Output : Number of that word present in the sentence : 2

Process

The word Home present in the sentence 2 times

Question 39

Write a Java program to find the first non-repeated character in a given string.

Here's an example of the expected input and output:

Input string: "minimum"

Output: "n"

Question 40

Find the first non repeated character substring

Arijit = Arij

Question 41

Count number of prime digit in a Number

Input : 254786135

Output : 5

Question 42

Count the sum of prime digits in a given number

Input: 2345678910

Output: 17

Explanation: 2+3+5+7 = 17

Question 43

Calculate the sum of all digits of a numbers. If the sum is greater than 9 then again sum the digits. The sum is continuing until the sum becomes one digit.

Input : 195

Output : 6

Explanation: $1+9+5 = 15 > 9$, so $1+5=6$

Question 44

Find average of a number array between two limit numbers (except the two numbers)

Input

5

1

2

3

4

5

1

4

Output

2

Question 45

Write a program that reads in a list of integers and prints out the smallest and largest integer in the list.

Your program should first ask the user to enter the number of integers they want to input. Then it should prompt the user to enter each integer one by one.

After all the integers have been entered, the program should output the smallest and largest integer in the list.

Input

Enter the number of integers: 5

Enter integer 1: 10

Enter integer 2: -5

Enter integer 3: 20

Enter integer 4: 0

Enter integer 5: 15

Output

Smallest integer: -5

Largest integer: 20

Question 46

Input : 1 2 4 8 9 11 12 14

Output : 3 5 6 7 10 13

Question 47

aabbbcc = a2b3c2
abbccc = a1b2c3

Question 48

a2b3c1 = aabbbc

Question 49

Write a Java program to implement a simple calculator. The program should prompt the user to enter two numbers and an operator (+, -, *, or /) and then perform the corresponding operation and display the result.

For example, if the user enters 4, 5, and +, the program should display 9 as the result. Similarly, if the user enters 10, 3, and *, the program should display 30 as the result.

Your program should handle invalid inputs gracefully, for example, if the user enters an operator that is not one of the four allowed operators or if the user enters non-numeric inputs.

Question 50

Input : 123Arijit1998Arijit10

Process :

Array : [123, Arijit, 1998, Arijit, 10]

Now you should sum up the position of that array where numbers are stored.

123 + 1998 = 2121

Output :

2131

Question 51

Input: "Hello@123World456"

Process:

Array: ["Hello", "@", "123", "World", "456"]

Now you should concatenate all the strings in the array and multiply it with the sum of the position of that array where numbers are stored.

Sum of positions of numbers in the array = 3 + 5 = 8

Concatenated string = "Hello@123World456"

Result = 8 * "Hello@123World456" =

"Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456"

Output:

"Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456Hello@123World456"

Hello@123World456"

Question 52

Intersection between two given set

Question 53

Books are identified by International Standard Book Number(ISBN). ISBNs contain 13 digits, arranged in five groups such as 978-0-13-110362-8. The first group (GSI prefix) is currently either 978 or 979. The group identifier specifies the language or country of origin (for example 0 and 1 are used for English speaking countries). The publisher code identifies the publisher (13 is the code for Prentice Hall, New Jersey). The item number is assigned by the publisher to identify a specific book (110362 is the code for the book "The C Programming Language" by Kernighan and Ritchie) and ISBN ends with a check digit that's used to verify the accuracy of preceding digits. Write a program that breaks down the ISBN entered by the user.:

Sample Input:

Enter ISBN: 978-0-13-110362-8

Sample Output:

GSI Prefix: 978

Group Identifier: 0

Publisher Code: 13

Item Number: 110362

Check Digit: 8

Note: The number of digits in each group may vary. You can't assume that the groups have the length shown in the sample output. You can test your program with actual ISBN values (usually found on the back cover of a book and on the copyright page)

Question 54

Find maximum prime from diagonals of a matrix

7 8 9

4 5 6

1 2 3

Max prime of two diagonal = 7

Question 55

12 4 8

5 7 6

4 8 13

Maxumum prime of the 2 diagonals 13

Question 56

Create a class Rectangle with the following attributes:

length - double

breadth - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - calculateArea and calculatePerimeter in Solution class.

calculateArea method:

This method will take a Rectangle object as input.

The method will calculate the area of the rectangle and return it as a double.

calculatePerimeter method:

This method will take a Rectangle object as input.

The method will calculate the perimeter of the rectangle and return it as a double.

These above mentioned static methods should be called from the main method.

For calculateArea method - The main method should print the area returned by the method.

For calculatePerimeter method - The main method should print the perimeter returned by the method.

Before calling these static methods in main, use Scanner object to read the values of two Rectangle objects referring attributes in the above mentioned attribute sequence.

Example Input:

5.5

3.2

7.0

4.5

Example Output:

17.6

17.4

31.5

23.0

Question 57

Find total interest of an amount.

Create a class Account with following attributes :

id - int

balance - double

interestRate - double

Take another integer as number of year.

After that calculate total interest using formula:

Percentage = interestRate * 100 / time

Total Interest = Percentage + interestRate

Question 58

Define two functions as given below.

1. validateScore: This function will validate a given number whether it is less than or equal to 100 and greater than 0.

If these conditions are matched, function will return True, otherwise return False

2. findValidScores : This function will take a list of scores (numbers) as input, and return a list of scores from the input

list after validation . Use the method validateScore to validate scores in the input list. If the input list does not contain

any valid score, function should return None.

Define the main section for the following:

1. Create a list of positive integers by taking user inputs

To create the list, do the following:

a. First read the number of elements (integer) you want to store in the list as input from user

b. Read a number as input and add to the list. Repeat this step for the number of elements to be stored in the list (considered in the first line of input)

2. Call the method findValidScores by passing the list created in point #1 as argument.

3. Print a message "Valid Scores =" (excluding the quotes) followed by the list returned by the method findValidScores in the same line. If the method returns None, print the message "No valid score found." (excluding the quotes).

Sample Input description:

The first input is the number of elements to be added to the list.

The next set of inputs are the numbers to be added to the list one after another and is repeated for the number of elements given in the first line of input.

Check the sample input and output below.

Sample Input 1:

For example, if you want to create a list of 5 positive numbers with values 10,20,101,90 and 0 then the input should be read/given as follows:

5
10
20
101
90
0

And the output for the above set of input will be:

Valid Scores = [10, 20, 90]

Sample Input 2:

8
-100
101
0
.80
101
120
-75
0.78

Output:

No valid score found.

Question 59

Create class Document with below attributes

```
id - int
title - String
folderName - String
pages - int
```

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - docsWithOddPages in Solution class.

This method will take array of Document objects and return another array with Document objects which has odd number of pages.

This method should be called from main method and display values of returned objects as shared in the sample (in ascending order of id attribute).

Before calling this method, use Scanner object to read values for four Document objects referring attributes in the above sequence.

Next call the method and display the result.

Consider below sample input and output:

Input:

```
1
resume
personal
50
2
question1
exams
55
3
question2
exams
45
4
India
misc
40
```

Output (each line has values separated by single space):

```
2 question1 exams 55
3 question2 exams 45
```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like nextInt(), nextDouble() etc.

Usually, this is not an issue, but this may be visible while calling nextLine() immediately after those methods.

Consider below input values:

```
22
```

hello

Referring below code:

```
Scanner sc = new Scanner(System.in);
int x = sc.nextInt();
String str = sc.nextLine(); -> here we expect str to have value hello. Instead it may
be "".
```

If above issue is observed, then it is suggested to add one more explicit call to `nextLine()` after reading numeric value.

Question 60

Create a Class Course with below attributes:

CourseName - String

CourseNumber - int

Mode - String

Sharedata - boolean

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - `findCourseStatus` in Solution class.

This method will take a String parameter along with the other parameter as array of Course objects.

The method will return status where the String parameter appears in the side attribute (with case insensitive search).

If the number of the course greater than equal to 1000, print "High", if greater or equal to 500 but less than 1000 then print "Medium" and if less than 500 then print "Low".

If the taken string not match with the Course name then print "No course found".

Consider below sample input and output:

Input 1:

Java

1200

Online

False

Python

600

Offline

False

HTML

800

Offline

False

JavaScript

300

Online

True

Python

Output 1:

Medium

=====

Input 2:

Java

1200

Online

False

Python

600

Offline

False

HTML

800

Offline

False

JavaScript

300

Online

True

PHP

Output 2:

No course found

Question 61

Create a Class Course with below attributes:

CourseId - Int

CourseName - String

CourseRating - double

Mode - String

HaveCertificate - boolean

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - CountCourse in Solution class.

This method will take a String parameter and a double parameter along with the other parameter as array of Course objects.

The method will return the total number of courses which have certificate facility and the rating of the course object is greater

than or equal to the given parameter with match the mode(with case insensitive search).

If no count is there then print "No course found".

Consider below sample input and output:

Input 1:

1001

Java

5

Online

True

1002

Python

```
3
Offline
False
1003
HTML
4
Offline
True
1004
JavaScript
2
Online
False
3
Online

Output 1:
-----
1
=====
Input 2:
-----
1001
Java
5
Online
True
1002
Python
3
Offline
False
1003
HTML
4
Offline
True
1004
JavaScript
2
Online
False
5
Offline

Output 2:
-----
No course found
```

Question 62

Create the class Flowers with below attribute:

```
flowerId-int;
flowerName-String;
```

```
price-int;  
rating-int;  
type-String;
```

The above attributes must be private, write getters and setters and parametrized constructor as well.

Create class MyClass with main method.

Implement one static method: findMinPriceByType in MyClass class.

findMinPriceByType Method:

This method will take two input parameters: Array of Flower objects and String parameter. This method will return the minimum

priced flower object from array of Flower objects for the given type(String parameter), only if the rating is greater than

3. If no Flowers with the above conditions are present in the array of the flower objects, then the method will return null.

For this method-The main method should print the flowerId from the returned object if the returned value is not null.

else it should print "There is no flower with given type".

input1:

123

Yellow trout lilly

3000

5

ephemerals

345

snowdrop

2500

4

ephemerals

321

Yarrow

1000

4

perennials

213

red trillium

2250

4

ephemerals

ephemerals

output:

213

input2:

123

Yellow trout lilly

3000

5

ephemerals
345
snowdrop
2500
4
ephemerals
321
Yarrow
1000
4
perennials
213
red trillium
2250
4
ephemerals
Biennial
output2:
There is no flower with given type.

Question 63

Create a class Fruits with below attributes as private

- int fruitId
- String fruitName
- int price
- int rating

Write getters, setters and parameterized constructor in the abovementioned attribute sequence as required.

Create class Solution with main method

Implement One static method-

- findMaximumPriceByRating

findMaximumPriceByRating in the solution class.

This method will take array of Fruit objects, int parameter and returns the maximum priced fruit

object from array of Fruit objects whichever has rating greater than the rating (rating parameter passed). This method should be called from main method.

Write code to perform following tasks:

1. Take necessary input variable and call findMaximumPriceByRating.

For this method- The main method should print the fruitId from returned object. As if the returned value is not null, or it should print "No such Fruit".

Note:- Before calling this static method, use Scanner object to read the values of four Fruits

objects. No two Fruits will have same price. All these searches should be case insensitive.

Consider below sample input and output to test your code.

Input

555

Apple

200

5

777

Orange

150

4

333

Banana

100

4

888

Avocado

250

4

3

Output

888

Question 64

Create a class Vegetable with below attributes:

VegetableId - int

VegetableName - String

Price - int

Rating - int

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findMinimumPriceByRating in Solution class.

findMinimumPriceByRating method:

The method will return minimum price object from array of vegetable objects, whichever by rating greater than given rating.

If no vegetable with the above condition is present in the array of vegetable object, then the mothod return null.

Note :

No vegetable has the same price.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findMinimumPriceByRating method - The main method should print the id of vegetable object or it should print

"No such Vegetables".

Eg: 1002

where 1002 is the id whose price is lowest

Before calling these static methods in main, use Scanner object to read the values of four Vegetable objects referring attributes in the above mentioned attribute sequence. Next, read the value of int parameter for rating.

Consider below sample input and output:

Input1

1001

Carrot

90

5

1002

Tomato

40

4

1003

Bectroot

80

4

1004

Onion

78

3

2

Output1

1002

=====

Input2

1001

Carrot

90

5

1002

Tomato

40

4

1003

Bectroot

80

4

1004

Onion

78

3

7

Output2

No such Vegetables

Question 65

Create a class Beach with the below attributes:

beachId - int

beachName - String

beachRating - int

beachCost - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findLeastRatingWithName in Solution class.

findLeastRatingWithName method:

This method will take three input parameters - array of Beach objects and string parameter and a int parameter. The method will return the minimum rating attribute from beach objects whose name is matched with string parameter and whose amount is greater than the int parameter. If no data found of beach objects, then the method should return 0.

Note : All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findLeastRatingWithName method - The main method should print the least rating from the beach array. If return 0, then you should print "No beach found".

Input ->

1001

Puri

3

8600

1002

Digha

5

6200

1003

Digha

3

4000

1004

Digha

4

5500

digha

5000

Output ->

4

Question 66

Create class Medicine with below attributes:

MedicineName - String

batch - String

disease - String

price - int

Create class Solution and implement static method "getPriceByDisease" in the Solution class.

This method will take array of Medicine objects and a disease String as parameters.

And will return another sorted array of Integer objects where the disease String matches with the

original array of Medicine object's disease attribute (case insensitive search).

Write necessary getters and setters.

Before calling "getPriceByDisease" method in the main method, read values for four Medicine objects

referring the attributes in above sequence along with a String disease. Then call the "getPriceByDisease"

method and print the result.

Input

dolo650

FAC124W

fever

100

paracetamol

PAC545B

bodypain

150

almox

ALM747S

fever

200

aspirin

ASP849Q

flu

250

fever

Output

100

200

Question 67

Create class Movie with below attributes:

movieName - String

company - String

genre - String

budget - int

Create class Solution and implement static method "getMovieByGenre" in the Solution class. This method will

take array of Movie objects and a searchGenre String as parameters. And will return another array of Movie

objects where the searchGenre String matches with the original array of Movie object's genre attribute

(case insensitive search).

Write necessary getters and setters.

Before calling "getMovieByGenre" method in the main method, read values for four Movie objects referring the

attributes in above sequence along with a String searchGenre. Then call the

"getMovieByGenre" method and write

logic in main method to print "High Budget Movie",if the movie budget attribute is greater than 800000000 else

print "Low Budget Movie".

Input

aaa

Marvel

Action

250000000

bbb

Marvel

Comedy

25000000

ccc

Marvel

Comedy

2000000

ddd

Marvel

Action

300000000

Action

Output

High Budget Movie

High Budget Movie

Question 68

Create class Inventory with below attributes:

inventoryId - String

maximumQuantity - int

currentQuantity - int

threshold - int

Create class Solution and implement static method "Replenish" in the Solution class.

This method will take array of Inventory objects and a limit int as parameters.

And will return another array of Inventory objects where the limit int is greater than or equal to the original array of

Inventory object's threshold attribute.

Write necessary getters and setters.

Before calling "Replenish" method in the main method, read values for four Inventory objects referring the attributes in above

sequence along with a int limit.Then call the "Replenish" method and write logic in main method to print "Critical Filling",

```
if the threshold attribute is greater than 75. Else if the threshold attribute is
between 50 and 75 then print
"Moderate Filling". Else print "Non-Critical Filling"
```

Input

```
1
100
50
50
2
200
60
40
3
150
35
45
4
80
45
40
45
```

Output

```
2 Non-Critical Filling
3 Non-Critical Filling
4 Non-Critical Filling
```

Question 69

Create a class Resort with the below attributes:

```
resortId - int
resortName - String
Category - String
resortPrice - double
resortRating - double
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findAvgPrice in Solution class.

findAvgPrice method:

This method will take two input parameters - array of Resort objects and a string parameter.

The method will return the average of the prices from resort objects whose Category matched with passed as parameter and rating will greater than 4.

If no resort present found, then the method should return 0.

Note : All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findAvgPrice method - The main method should print the average price as it is if the returned value is greater than 0, or it should print "No such Resort found".

Before calling these static methods in main, use Scanner object to read the values of four resort objects referring attributes in the above mentioned attribute sequence. Next, read the value for category.

Input

1004

ABC

3 star

5400.00

4.5

1001

DEF

5 star

4781.45

4.0

1002

GHI

3 star

6235.72

4.2

1008

JKL

2 star

7454.00

4.2

3 star

Output

The average price of 3 star:5817

Question 70

Create class Team with below attributes:

tId - int

tName - String

tCountry - String

tRun - int

The above attributes should be private. Write Getter and Setter and parametrized constructor as required.

Create class Solution with main method.

Implement One static method-

● findPlayer

findPlayer in the solution class.

This method will take array of Team objects, int parameter run and String country and returns the Team object where taken country is matched with the country of team object and team run is greater than the taken run.

Write code to perform following tasks:

1. In main mehod, take 1st input as integer parameter which shows the number of team object.
2. Take necessary input variable and call findPlayer.
3. All searches should be case insensitive.

For this method- The main method should print the team object. As if the returned value is not null,
or it should print "No team is found from the given country and run".

Input

```
3
1
Arijit
India
43
2
Doremon
China
40
3
Donal
England
32
35
China
```

Output

```
2
Doremon
China
40
```

Question 71

Create a class Motel with the below attributes:

```
motelId - int
motelName - String
dateOfBooking - String (in the format dd-mon-yyyy)
noOfRoomsBooked - int
cabFacility - String
totalBill- double
```

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The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement one static method - totalNoOfRoomsBooked in Solution class.

totalNoOfRoomsBooked method:

This method will take two input parameter - array of Motel objects and a String parameter.

The method will return the total numbers of rooms booked from array of Motel objects if the cab facility

attribute matches with the given String parameter(cab facility) and the number of rooms booked is greater than 5.

If no rooms are booked with the above criteria in the array of Motel objects, then the method should return 0.

Note :

No two Motel object would have the same motelId.

dateOfBooking is stored in the format dd-mon-yyyy(eg. 01-Jan-2022)

The above mentioned static method should be called from the main method.

For totalNoOfRoomsBooked method - The main method should print the total number of booked rooms

as it is, if the returned value is greater than 0, else it should print "No such rooms booked"

Before calling these static methods in main, use Scanner object to read the values of four Motel objects

referring attributes in the above mentioned attribute sequence.

Next, read the value of one String parameter for capturing the cab facility

Input

1001
M&M
01-Dec-2022
5
Yes
30000
1002
BestStay
10-Jan-2022
3
Yes
27000
1003
Novatel
11-Jun-2022
5
Yes
25000
1004
Chola
01-Sep-2022
7
Yes
72000
Yes

OutPut

7

Question 72

Create class Associate with below attributes:

```
id - int
name - String
technology - String
experienceInYears - int
```

Create class Solution and implement static method "associatesForGivenTechnology" in the Solution class.

This method will take array of Associate objects and a searchTechnology String as parameters.

And will return another array of Associate objects where the searchTechnology String matches with the original array of

Associate object's technology attribute (case insensitive search) and experienceInYears attribute should be multiples of 5.

Write necessary getters and setters.

Before calling "associatesForGivenTechnology" method in the main method, read values for five associate objects referring

the attributes in above sequence along with a String searchTechnology.

Then call the "associatesForGivenTechnology" method and write the logic to print the id's in the main method.

Input

```
101
Alex
Java
15
102
Albert
Unix
20
103
Alferd
Testing
13
104
Alfa
Java
15
105
Almas
Java
29
Java
Output
```

```
101
104
```

Question 73

Create class Player2 with below attributes:

```
id - int
name - String
iccRank - int
```

```
matchesPlayed - int
```

```
runsScored - int
```

Create class Solution and implement static method "findAverageOfRuns" in the Solution class.

This method will take array of Player objects and a target int as parameters.

And will return another double array where the target int is lesser than or equal to the original array of Player object's

matchesPlayed attribute and contains the average run scored by each player satisfying above condition.

Write necessary getters and setters.

Before calling "findAverageOfRuns" method in the main method, read values for four Player objects referring the attributes

in above sequence along with a int target.

Then call the "findAverageOfRuns" method and write logic in main method to print "Grade A",if the calculated averageRun

value is 80 to 100. Else if the averageRun value is between 50 and 79 then print "Grade B". Else print "Grade C".

Input

100

Sachin

5

150

13000

101

Sehwag

4

120

10000

103

Dhoni

7

110

7000

104

Kohli

15

57

4400

100

Output

Grade A

Grade A

Grade B

Question 74

Create a Class Player with below attributes:

```
id - int
```

```
country - String
```

```
side - String
```

price - double

Write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement static method - searchPlayerForMatch in Solution class.

This method will take a String parameter along with the other parameter as array of Player objects.

The method will return array of Player where the String parameter appears in the side attribute (with case insensitive search).

This method should be called from main method and display the id of returned objects in ascending order.

Before calling this method(searchPlayerForMatch) in the main method, use Scanner object to read values for four Player objects

referring the attributes in the above sequence, then, read the value for search parameter.

Next call the method searchPlayerForMatch, write the logic to sort the id in ascending order (in main method) and display the result.

Consider below sample input and output:

Input:

1

India

Batting

2500000

2

Australia

Batting

1000000

3

Srilanka

Bowling

1700000

4

England

Bowling

2000000

Batting

Output:

1

2

Question 75

Create the class RRT(Rapid response team) with the below attributes:

ticketNo-int

raisedBy-String

assignedTo-String

priority-int

project-String

All attributes should be private, write getters and setters and parameterized constructor as required.

Create class MyClass with main method.

Implement a static method-getHighestPriorityTicket in MyClass class.

getHighestPriorityTicket method:

This method will take an array of RRT objects ,and a String value as parameters. This method will return the RRT object with highest priority ticket from the array of the RRT objects for the given project(String parameter passed). Highest priority is the one which has lesser value.

for example: 1 is considered as high priority and 5 is considered as low priority.

If no RRT with the above condition is present in the array of the RRT objects, then the method should return null.

The main method should print the ticketNo, raisedBy and assignedTo from returned object if the returned object is not null. If the returned object is null then main method should print "No such Ticket".

input1:

123

Velantish

Mani

3

Xperience

234

Sathish

Akshaya

1

AIG

345

John

Jack

2

AIG

456

Bhuvi

Jack

5

AIG

AIG

output1:

234

Sathish

Akshaya

Input2:

123

Velantish

```
Mani
3
Xperience
234
Sathish
Akshaya
1
AIG
345
John
Jack
2
AIG
456
Bhuvi
Jack
5
AIG
Xplore

output2:
-----
No such ticket.
```

Question 76

Group the employees by city
Input:

Employee | City

Arijit Kolkata
Minaz Mumbai
Rahul Chennai
Rajesh Kolkata
Sekhar Chennai
Kasim Chennai
Output:

Kolkata: Arijit,Rajesh
Mumbai: Minaz
Chennai: Rahul,Sekhar,Kasim

Question 77

Create a class Employee with the following attributes:
name (string)
age (int)
salary (double)
Write getters, setters, and a parameterized constructor in the above-mentioned attribute sequence as required.
Create a class Solution with the main method.

Implement two static methods - calculateYearlySalary and calculateTax in the Solution class.

calculateYearlySalary method:

This method will take an Employee object as input.

The method will calculate the yearly salary of the employee (assuming that the employee works for 12 months in a year) and return it as a double.

calculateTax method:

This method will take an Employee object as input.

The method will calculate the tax to be paid by the employee based on the following rules:

If the yearly salary is less than or equal to 50000, the tax is 10% of the yearly salary.

If the yearly salary is greater than 50000 but less than or equal to 100000, the tax is 20% of the amount over 50000 plus 10% of the first 50000.

If the yearly salary is greater than 100000, the tax is 30% of the amount over 100000 plus 20% of the amount between 50000 and 100000 plus 10% of the first 50000.

The method will return the tax amount as a double.

These above-mentioned static methods should be called from the main method.

For calculateYearlySalary method - The main method should print the yearly salary returned by the method.

For calculateTax method - The main method should print the tax amount returned by the method.

Before calling these static methods in main, use a Scanner object to read the values of an Employee object's attributes.

Example Input:

John

30

55000.0

Example Output:

Yearly salary of John: 660000.0

Tax to be paid by John: 183000.0

Question 78

Create a class Car with the following attributes:

make - String

model - String

year - int

price - double

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - findMostExpensiveCar and getCarByMakeAndModel in Solution class.

findMostExpensiveCar method:

This method will take an array of Car objects as input.

The method will return the Car object with the highest price.

If the array is empty, the method should return null.

getCarByMakeAndModel method:

This method will take three input parameters - array of Car objects, make and model (both strings).

The method will return the Car object that matches the make and model attributes passed as parameters.

If no car with the given make and model is present in the array of car objects, then the method should return null.

Note: The search should be case insensitive.

These above mentioned static methods should be called from the main method.

For findMostExpensiveCar method - The main method should print the make, model, year and price of the returned Car object.

If the returned value is null then it should print "Sorry - No car is available".

For getCarByMakeAndModel method - The main method should print the year and price of the returned Car object. If the returned value is null then it should print "Sorry - No car is available".

Before calling these static methods in main, use Scanner object to read the values of four Car objects referring attributes in the above mentioned attribute sequence. Next, read the values for make and model.

Input:

Mercedes

S-Class

2020

1250000

BMW

5-Series

2019

900000

Audi

A4

2022

800000

Jaguar

F-PACE

2021

1100000

Audi

A4

Output:

Mercedes

S-Class

1250000.0

2020

2022

800000.0

Question 79

Create a class Book with below attributes

id - int

pages - int

title - String

author - String

price - double

The above attributes should be private, write getters and setters and parameterized constructor as required.

Create a class Solution with main method

Implement two static methods - findBookWithMaximumPrice and searchBookByTitle in Solution class.

findBookWithMaximumPrice :

=====

Create a static method findBookWithMaximumPrice in the Solution class. This method will take array of Book objects and returns the Book object having the maximum Price if found else return null if not found.

searchBookByTitle :

=====

Create a static method searchBookByTitle in the Solution class. This method will take array of Book objects and Title as input and returns the Book object having the mentioned Title if found else return null if not found.

These methods should be called from the main method.

write code to perform the following tasks.

1. Take necessary input variable and call findBookWithMaximumPrice. For this method - The main method should print the Book object with the maximum of mentioned attribute as it is if the returned value is not null, or it should print "No Book found with mentioned attribute."

2. Take necessary input variable and call searchBookByTitle. For this method - The main method should print the Book object details as it is, if the returned value is not null or it should print "No Book found with mentioned attribute."

The above mentioned static methods should be called from the main method. Also write the code for accepting the inputs and printing the outputs. Don't use any static test or formatting for printing the result. Just invoke the method and print the result.

All String comparison needs to be in case sensitive.

Input:

1

845

Bengali

Arijit
525.50
2
456
English
Raju
412.30
3
1022
History
Kaka
525.50
4
125
geography
MN
524
English
Output:
1 Bengali
3 History
2
456

Question 80

Create a class Phone with below attributes:

phoneId - int
os - String
brand - String
price - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with main method.

Implement two static methods - findPriceForGivenBrand and getPhoneIdBasedOnOs in Solution class.

findPriceForGivenBrand method:

This method will take two input parameters - array of Phone objects and string parameter brand. The method will return the sum of the price attribute from phone objects for the brand passed

as parameter. If no phones with the given brand is present in the array of phone objects, then the method should return 0.

getPhoneIdBasedOnOs method:

This method will take a String parameter os, along with the array of Phone objects.

The method will return the phone object, if the input String parameter matches with the os attribute of the phone

object and its price attribute is greater than or equal to 50000. If any of the conditions are not met, then the

method should return null.

Note : No phone object would have the same value for os attribute. All phone object would have the price greater than 0.

All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findPriceForGivenBrand method - The main method should print the price as it is if the returned price is greater than 0, or it should print "The given Brand is not available".

For getPhoneIdBasedOnOs method - The main method should print the phoneId of the returned phone object. If the returned value is null then it should print "No phones are available with specified os and price range".

Before calling these static methods in main, use Scanner object to read the values of four Phone objects referring attributes in the above mentioned attribute sequence. Next, read the value for brand and os.

Input

111

iOS

Apple

30000

222

android

Samsung

50000

333

Symbian

HTC

12000

444

Paranoid

HTC

89000

Blackberry

aNdROID

Output

The given Brand is not available

222

Question 81

Create a class Player with below attributes:

playerId - int

skill - String

level - String

points - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findPointsForGivenSkill and getPlayerBasedOnLevel in Solution class.

findPointsForGivenSkill method:

This method will take two input parameters - array of Player objects and string parameter skill. The method will return the sum of the points attribute from player objects for the skill passed as parameter. If no player with the given skill is present in the array of player objects, then the method should return 0.

getPlayerBasedOnLevel method:

This method will take two String parameters level and skill, along with the array of Player objects.

The method will return the player object, if the input String parameters matches with the level and

skill attribute of the player object and its point attribute is greater than or equal to 20. If any

of the conditions are not met, then the method should return null.

Note : No two player object would have the same value for skill and level attribute. All player object

would have the points greater than 0. All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findPointsForGivenSkill method - The main method should print the points as it is if the returned

price is greater than 0, or it should print "The given Skill is not available".

For getPlayerBasedOnLevel method - The main method should print the playerId of the returned player object.

If the returned value is null then it should print "No player is available with specified level, skill and eligibility points".

Before calling these static methods in main, use Scanner object to read the values of four Player objects

referring attributes in the above mentioned attribute sequence. Next, read the value for skill and level.

Please consider the skill value read above as skill parameter for both the static methods.

Input:

101

Cricket

Basic

20

102

Cricket

Intermediate

25

111

Football

Intermediate

50


```

113
BaseBall
Advanced
100
Cricket
Intermediate
Output:
-----
45
102

```

Question 82

Create a class Cinema with the below attributes:

```

movieId - int
director - String
rating - int
budget - int

```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findAvgBudgetByDirector and getMovieByRatingBudget in Solution class.

findAvgBudgetByDirector method:

```
-----
```

This method will take two input parameters - array of Movie objects and string parameter director.

The method will return the average of the budget attribute from Movie objects directed by the director passed as parameter.

If no movie with the given director is present in the array of movie objects, then the method should return 0.

getMovieByRatingBudget method:

```
-----
```

This method will take two int parameters rating and budget, along with the array of movie objects.

The method will return the movie object, if the input parameters rating and budget, matches with the rating and budget attribute of the movie object respectively.

Also check if rating is a factor of budget (eg: 3 is a factor of 12 because 3 divides 12 without leaving a remainder).

If any of the conditions are not met, then the method should return null.

Note : No two movie object would have the same value for rating and budget attributes. All the searches should be case insensitive.

The budget mentioned are in crores and in INR.

These above mentioned static methods should be called from the main method.

For findAvgBudgetByDirector method - The main method should print the average budget as it is if the returned value is

greater than 0, or it should print "Sorry - The given director has not yet directed any movie".

For getMovieByRatingBudget method - The main method should print the movieId of the returned movie object. If the returned value is null then it should print "Sorry - No movie is available with the specified rating and budget requirement".

Before calling these static methods in main, use Scanner object to read the values of four Movie objects referring attributes in the above mentioned attribute sequence. Next, read the value for director, rating and budget.

Input

1101

GVM

4

100

1201

Shankar

5

500

1301

Shankar

3

50

1401

GVM

5

300

GVM

5

300

Output

200

1401

Question 83

Create class Sim2 with below attributes:

id - int

company - String

balance - int

ratePerSecond - double

circle - String

Create class Solution and implement static method "matchAndSort" in the Solution class. This method will take array of Sim objects, search_circle String and search_rate double as parameters.

And will return another Sim array where the search_circle matches with the circle parameter of the original Sim array and

the search_rate double is greater than the original array of Sim object's ratePerSecond attribute which is also sorted by means of balance attribute in descending order.

Write necessary getters and setters.

Before calling "matchAndSort" method in the main method, read values for four Sim objects referring the attributes in above sequence along with a String search_circle and a double search_rate. Then call the "matchAndSort" method and write logic in main method to print the id's of the result obtained.

Input

```
1
jio
430
1.32
mumbai
2
idea
320
2.26
mumbai
3
airtel
500
2.54
mumbai
4
vodafone
640
3.21
mumbai
mumbai
3.4
```

Output

```
4
3
1
2
```

Question 84

Create class Player3 with below attributes:

```
id - int
name - String
Country - String
matchesPlayed - int
runsScored - int
```

the above attribute should be private ,write getter and setter and parametrized constructor as well.

Create class Solution and implement static method "findPlayerName" in the Solution class.

This method will take array of Player objects and a String as country parameters and int as run parameter.

And will return another array of Player object where the String is matched with country and run scored of that Player object

is greater than the input int parameter. All the searches should be case insensitive. Before calling "findPlayerName" method in the main method, read values for Player objects referring the attributes in above sequence along with a int target. Then call the "findPlayerName" method and write logic in main method to print Player object. If any of the input doesn't match with the following attributes then print "No player found".

Note:

1. First take input as the number of player object, then the player details
2. Show the output as the dictionary order of the players name
3. First print player id, and then use ' : ' sign and after that player name

3 : Arijit
1 : Minaz
5 : Priyanka

Input : 1

5
1
Arnab
India
4
94
2
Virat
UK
5
68
3
Arijit
India
82
87
4
Raj
India
21
78
5
Priyanka
India
12
86
India
80
Output : 1

3:Arijit
1:Arnab
5:Priyanka

Input : 2

```
-----  
5  
1  
Arnab  
India  
4  
94  
2  
Virat  
UK  
5  
68  
3  
Arijit  
India  
82  
87  
4  
Raj  
India  
21  
78  
5  
Priyanka  
India  
12  
86  
UK  
80  
  
Output : 2  
-----  
No player found
```

Question 85

Create a class BankAccount with the following attributes:

```
accountNumber - int  
accountHolderName - String  
balance - double
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - withdraw and deposit in Solution class.

withdraw method:

```
-----  
This method will take two input parameters - array of BankAccount objects and the  
account number (int) from which withdrawal  
should be done.
```

The method will check if the account number is present in the array of BankAccount objects.

If present, the method will then check if the balance is sufficient to withdraw the requested amount (double). If yes, it will

deduct the amount from the account balance and return the updated balance.
If the account number is not present or the balance is not sufficient, the method should return -1.

deposit method:

This method will take two input parameters - array of BankAccount objects and the account number (int) in which deposit should be done.

The method will check if the account number is present in the array of BankAccount objects.

If present, it will add the deposit amount (double) to the account balance and return the updated balance.

If the account number is not present, the method should return -1.

These above mentioned static methods should be called from the main method.

For withdraw method - The main method should print the updated balance if the returned value is greater than or equal to 0, or

it should print "Sorry - Insufficient balance" if the returned value is -1, or it should print "Sorry - Account not found" if

the returned value is -2.

For deposit method - The main method should print the updated balance if the returned value is greater than or equal to 0, or

it should print "Sorry - Account not found" if the returned value is -1.

Before calling these static methods in main, a parameterized constructor in the above mentioned attribute sequence as required.

Input

1001

Alice

5000.0

1002

Bob

10000.0

1003

Charlie

15000.0

1002

5000.0

1001

10000.0

Output

5000.0

15000.0

Question 86

Create a class Student with the following attributes:

name - String

rollNo - String

age - int

gender - char ('M' for male and 'F' for female)

```
course - String
semester - int
GPA - double
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - calculateAverageGPA and getStudentsByCourse in Solution class.

calculateAverageGPA method:

This method will take an array of Student objects as input.

The method will return the average GPA of all the students in the array.

If the array is empty, the method should return 0.

getStudentsByCourse method:

This method will take two input parameters - array of Student objects and a course (string).

The method will return an array of Student objects that are enrolled in the specified course.

If no student with the given course is present in the array of Student objects, then the method should return an empty array.

These above mentioned static methods should be called from the main method.

For calculateAverageGPA method - The main method should print the average GPA returned by the method.

If the returned value is 0, then it should print "Sorry - No students are available".

For getStudentsByCourse method - The main method should print the name, roll number and GPA of the returned Student objects.

If the returned value is an empty array, then it should print "Sorry - No students are available for the given course".

Before calling these static methods in main, use Scanner object to read the values of four Student objects referring attributes in the above mentioned attribute sequence. Next, read the value for a course.

Example Input:

```
-----
John
1234
20
M
CSE
2
3.5
Alice
5678
19
F
EEE
1
3.7
Bob
9101
21
M
```

CSE
4
3.9
Sarah
2345
20
F
ME
3
3.2
CSE

Example Output:

3.6
John
1234
3.5
Bob
9101
3.9

Question 87

Create a class College with the below attributes.

id-int
name -String
contactNo-int
address-String
pinCode-int

Write the getters and setters and parametrized constructor in the above mentioned sequence as required.

Create the class Solution with the main method.

Implement the two static methods:

- 1.findCollegeWithMaximumPincode
- 2.searchCollegeByAddress

findCollegeWithMaximumPincode method:

Create the Static method in the Solution Class.

This method will take array of the College objects and return the College object having maximum pincode if found else return null if not found.

for this method ,main method will print College object with maximum pincode if the returned value is not null.if the returned value is null , then the main method will print "No college found with mentioned attribute".

searchCollegeByAddress method:

Create the Static method in the Solution Class.

This method will take array of College objects as input and address as input and return College object having the mentioned address if found else return null if not found.

for this method main method will print College object details as it is , if the returned value is not null. if the returned value is null then , main method will print "No college found with mentioned attribute".

input 1:

4
109
ACT
2500256
mumbai
695001
107
MCE
2500254
malapuram
612354
113
CTE
2500252
chennai
623145
102
SCT
2500255
AP
523641
AP

OUTPUT1:

id-109
name-ACT
contactNo-2500256
address-mumbai
pincode-695001
id-102
name-SCT
contactNo-2500255
address-AP
pincode-523641

INPUT2:

4
111
MJT
2500251

```
Calicut
401235
105
MET
2500256
kochi
668745
115
IIT
2500262
banglore
569874
110
ACT
2500263
delhi
687945
delhi
```

```
output2:
id-110
name-ACT
contactNo-2500263
address-delhi
pincode-687945
id-110
name-ACT
contactNo-2500263
address-delhi
pincode-687945
```

Question 88

Create a class newspaper with below attributes.

```
regNo-int
name-String
publicationYear-int
price-int
```

Write parametrised constructor as required.

Create class Solution with main method.

Implement 2 static methods findTotalPriceByPublicationYear and searchNewspaperByName in solution class.

findTotalPriceByPublicationYear method:

This method will take array of the Newspaper objects and int parameter type. This method will return the total price of newspapers from array of Newspaper objects, If the publication year attribute matches with the int parameter passed. If no Newspaper with the given publication year is present

in the array ,then the method should return 0.

The main method should print total price if the returned value is greater than 0.Else it should print "No Newspaper found with the mentioned attribute".

searchNewspaperByName method :

This method will take the array of the Newspaper objects and the String parameter type.This method will return the Newspaper object from the Array of Newspaper objects,If the name attribute matches with the string parameter passed.

If no Newspaper with the given name is present in the array ,then the method should return null.

The main method should print the regNo,name,year of publication,and price if the returned Newspaper object is not null.
else it should print "No Newspaper found with the given name."

The above mentioned static methods should be called from main method.

input1:

134
Evening Mail
2018
79
178
Morning Mail
2017
65
176
Daily News
2017
98
219
Times of India
1988
15
2017
Times of India

output1:

163
regNo-219
name-Times of India
publicationYear-1988
price-15

input2:

201
Afternoon Mail
2019
79
202
Morning Mail
2014
65
156
Daily News
2004
98
278
Times of India
1967
15
2017
Times of America

output2:

No Newspaper found with the mentioned attribute

No Newspaper found with the given name

Question 89

Create a class Engine with the below attributes:

engineId - int

engineName - String

engine Type - String

enginePrice - double

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class MyClass with main method.

Implement two static methods - findAvgEnginePriceByType and searchEngineByName in MyClass class.

findAvgEnginePriceByType method: This method will take two input parameters Engine objects and String parameter.

The method will return the average price of Engines from array of Engine objects for the given engine type

(String parameter passed). If no Engine with the given type is present in the array of Engine objects, then the method should return 0.

searchEngineByName method: This method will take a String parameter and an array of Engine objects as the other parameter.

The method will return Engine object array in an ascending order of their engine ids, from the array of Engine objects whose

name attribute matches with the given engine name (String parameter passed). If no Engine with the given name is present in the array of Engine objects, then the method should return null.

Note: No two Engine object would have the same engine id.

All searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findAvgEnginePriceByType method: The main method should print the average enginePrice of Engines as it is, if the returned value is greater than 0, or it should print "There are no engine with given type".

For searchEngineByName method: The main method should print the engineId from the returned Engine object array if the returned value is not null. If the returned value is null, then it should print "There are no engine with the given name".

Before calling these static methods in main, use Scanner object to read the values of four Engine objects referring attributes in the above mentioned attribute sequence. next, read the value of two String parameters for capturing engine type and engine name respectively.

Input :

```
1001
Maruti
Diesel
20000
1002
Kia
Petro
17000
1003
Hyundai
Diesel
24000
1000
Maruti
Petrol
27500
Petrol
Maruti
```

Output :

```
27500
1000
1001
```

Question 90

create a class Antenna with below attributes.

```
antennaId=int
antennaName=String
projectLead=String
antennaVSWR=double
```

the above attribute should be private ,write getter and setter and parametrized constructor as well.

create class MyClass with main method.

implement two static methods searchAntennaByName and sortAntennaByVSWR in MyClass class.

searchAntennaByName :

This method will take an array of Antenna objects and the string value as input parameter.The method will find out Antenna name(String parameterpassed).

It will return Antennaid if found.if there is no Antenna that matches then the method will return zero.

The main method should print the antennaid,if the returned value is not 0.if the returned value is 0 then print,"There is no antenna with the given parameter".

sortAntennaByVSWR:

This method will take an array of Antenna Objects and a double value as input.

This method will return an array of Antenna objects sorted in ascending orderof their antennaVSWR, which is less than VSWR(double value passed).

The main method should print name of the project leads from the returned array if the returned value is not null.if the returned value is null then main method should print "No Antenna found";

input1:

111

Reconfigurable

Hema

0.4

222

Wearable

Kavya

0.9

333

Microstrip

Teju

0.3

444

Dielectric

Sai

0.65

Microstrip

0.5

output:

333

Teju

Hema

input 2:

111

Reconfigurable

Hema

0.3

222

Wearable

Kavya
0.9
333
Microstrip
Teju
0.4
444
Dielectric
Sai
0.65
Resonator
0.25
output:
There is no antenna with given parameter
No Antenna found

Question 91

create the class Song with below attributes.

songId - int
title - String
artist - String
duration-double

The above attributes should be private, getters, setters and parameterized constructor as required.

Create class MyClass with main method. Implement two static methods -
findSongDurationForArtist and
getSongsInAscendingOrder in MyClass class.

findSongDurationForArtist method:

This method will take two input parameters of Song objects and String parameter.

The method will return the sum of song duration from array of Song object for the given artist (String parameter passed).

If no Song with the given artist is present in the array of Song objects, then the method should return zero.

getSongsInAscendingOrder method:

This method will take input parameters array, of Song objects and String parameter. The method will return Song objects array

in an ascending order of their duration, from the array of Song objects whose artist attribute matches with the input String parameter.

If no Song with the given artist is present in the array of Song objects, then the method should return null.

Note:

1. No two Song object would have the same songid.
2. Combination of artist and duration should be unique.
3. All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For getfindSongDuration ForArtist method - in method should print the sum of duration of artist as it is, if the returned value is greater than 0 or it should print "There are no songs with given artist".

For getSongsinAscendingOrder method - The main method should print the song id and title from the returned song object array if the returned value is not null. If the returned value is null then it should print "There are no songs with given artist".

For Example,

112

ABC

where 112 is song ld. ABC is song title.

Before calling these static methods in main, use. Scanner object to read the values of five Song objects referring attributes in the above mentioned attribute sequence. Next, read two String values for capturing artists.

Sample Input 1:

2150

In time

Justin Timberlake

4

250

Cry Me

Justin Timberlake

3

1200

Mirrors

Justin Timberlake

5

1300

That's the way it is

celion dion

5

500

Ashes

celion dion

3

celion dion

Justin Timberlake

Sample output1:

8.0

250

Cry Me

2150

In time

1200

Mirrors

Sample Input 2:

2150
Cry Me
Justin Timberlake
3
1000
Why Not
Enrique Iglesias
5
1200
Mirrors
Justin Timberlake
5
1300
That's the way it is
Celion Dion
5
500
Ashes
celion Dion
3
Bryan Adams
Michael Larkson

Output 2:

There are no songs with given artist
There are no songs with given artist

Question 92

Create a class Music with below attributes:

playListNo - int
type -String
count - int
duration(minutes) -double.

The above attributes should be private. Write Getter and Setter and parametrized constructor as required.

Create class Solution with main method.

Implement two static methods: findAvgOfCount and sortTypeByDuration in Solution Class.

findAvgOfCount Method:

This method will take an array of Music objects and an int value as input parameters. The method will find out Average (as int) of count for those objects whose count is more than the given Count(int parameter passed). This method will return average if found. If there is no Type that matches then the method should return 0. for this method- main method should print average ,if the returned value is not 0. If the returned value is 0 then main method

should print "No playlist found".

sortByDuration method:

This method will take an array of Music objects and an int value as input parameters. This method should return an array of

Music objects in an ascending order of their duration which are more than the given duration (int parameter passed) value.

If there are no such objects, then the method should return null.

for this method - The main method should print the type from the returned Music object array if the returned value is not null.

If the returned value is null then the main method should print "No playlist found with mentioned attribute".

Consider the below input and output:

input1:

111

WorkOut

10

15.2

321

Dance Party

20

55.500

721

Childhood Jams

6

50.60

521

Chill

30

78.89

15

20

output1:

25

Childhood Jams

Dance Party

Chill

input2:

111

Oldies but Goodies

17

55

321

Guilty Pleasures

27

27

721

night drive

21

345

521

Rainy day

34

21

50

5000

output 2:

No playlist found.

No playlist found with mentioned attribute.

Question 93

Create a class Car with the below attributes:

carId - int

carName - String

fuelType - String

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class RemoveCar with main method.

Implement two static methods - RemoveAndRearrange in RemoveCar class.

RemoveAndRearrange method:

This method will take int input parameter. The method will return the new array of Car object and remove that car whose id is match with the input parameter. Remove the car details and return the array with continuous new ids.

If no car with the given id is present in the array of Car objects, then the method should return null.

Note: No two Car object would have the same engine id.

The above mentioned static methods should be called from the main method.

For RemoveAndRearrange method: The main method should print the id and car name as it is, if the returned value is not null, or it should print "There are no car with given id".

Before calling these static methods in main, use Scanner object to read the values of five objects referring attributes in the above mentioned attribute sequence. next, read the value of int parameters for capturing car id.

Input :

1001

Maruti

Diesel

1002

Kia

Petro

1003

Hyundai

Diesel

```
1004
Taxi
Petrol
1005
Truck
Diesel
1003
Output :
1001 : Maruti
1002 : Kia
1003 : Hyundai
1004 : Taxi
```

Question 94

Create a class MobileDetails with the below attributes:

```
MobileId - int
Price - int
Brand - String
isFlagShip - boolean
```

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class MobileDetails with main method.

Implement two static methods - getTotalPrice() and getSecondMin() in MobileDetails class.

getTotalPrice method:

This method will take two int input parameters which take object array and string brand. The method will return the total price of mobiles of that brand.

If no mobile with the given brand is present in the array of MobileDetails objects, then the method should return 0.

getSecondMin method:

This method will take object array which will return the second minimum price from the array and name of that mobile brand.

The above mentioned static methods should be called from the main method.

For getTotalPrice method: The main method should print the total price as it is, if the returned

value is not null, or it should print "There are no mobile with given brand".

Before calling these static methods in main, use Scanner object to read the values of five objects referring attributes in the above mentioned attribute sequence.

Input :

```
101
25000
Samsung
True
102
```

```

19000
Redmi
False
103
28000
Nokia
True
104
25000
Realme
False
105
40000
Samsung
True
Samsung
Output :
-----
65000
Samsung : 25000
Realme : 25000

```

Question 95

Create a class HeadSets with below attributes:

```

headsetName - String
brand-String
price - int
available- boolean

```

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findTotalPriceForGivenBrand and findAvailableHeadsetWithSecondMinPrice in Solution class.

findTotalPriceForGivenBrand method:

```
-----
```

The method will return the total price of Headsets from array of Headset objects for the given brand(String parameter passed).

If no Headsets with the given brand is present in the array of Headsets objects, then the method should return 0.

findAvailableHeadsetWithSecondMinPrice method:

```
-----
```

This method will take array of Headset objects as an input parameter and returns the available Headset object with the second lowest/minimum price from the given array of objects.

If no Headsets with the above condition is present in the array of Headsets objects, then the method should return null.

Note :

No two Headsets will have the same price.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findTotalPriceForGivenBrand method - The main method should print the returned total price as it is if the returned value is greater than 0 or it should print "No Headsets available with the given brand".

Eg: 4500

where 4500 is the total Headsets price

For findAvailableHeadsetWithSecondMinPrice method - The main method should print the headsetsName and price from the returned Headsets object if the returned value is not null.

If the returned value is null then it should print "No Headsets available".

Eg:Logitech Wired

1500

where Logitech Wired is the headsetName and 1500 is the price

Before calling these static methods in main, use Scanner object to read the values of four Headset objects referring attributes in the above mentioned attribute sequence.

Next, read the value of String parameter for capturing brand.

Consider below sample input and output:

Input1:

boAt BassHeads

boAt

1220

true

Over Ear Wired

boAt

549

true

In Ear with Mic

JBL

450

true

Buds 2 Neo

RealMe

500

true

boAt

Output:

1769

Buds 2 Neo

500

Input2:

boAt BassHeads

boAt

1220

false

Over Ear Wired

boAt

1549

false

In Ear with Mic

JBL

450

false

Buds 2 Neo

RealMe

500

false

boAt

Output:

2769

No Headsets available

Sample code snippet for reference:

Please use below code to build your solution.

```
import java.util.Scanner;
public class MyClass
{
    public static void main(String[] args)
    {
        //code to read values
        //code to call required method
        //code to display the result
    }
    public static int findTotalPriceForGivenBrand (/* required parameters to be added */)
    {
        Restricted for circulation outside TCS Xplore 9
        //method logic
    }
    public static HeadSets findAvailableHeadsetWithSecondMinPrice (/* required parameters
    to be
    added */)
    //method logic
    }
    class HeadSets
    {
        //code to build the class
    }
}
```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like nextInt(), nextDouble() etc.

Usually, this is not an issue, but this may be visible while calling nextLine() immediately after those methods.

Consider below input values:

1001

Savings

Referring below code:

```
Scanner sc = new Scanner(System.in);
```

```
int x = sc.nextInt();
```

String str = sc.nextLine(); -> here we expect str to have value Savings. Instead it may be "".

If above issue is observed, then it is suggested to add one more explicit call to nextLine() after reading numeric value.

Question 96

Create a class called Student with the below attributes:

rollNo - int
name - String
branch - String
score - double
dayScholar - boolean

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods -findCountOfDayscholarStudents and findStudentwithSecondHighestScore in Solution class.

findCountOfDayscholarStudents:

This method will take an array of Student objects as an input parameter . This method will calculate and return the count of Students whose score is greater than 80 and who are all from dayScholar. If no Student scored greater than 80 and from dayScholar are present in the array of Student objects, then the method should return 0.

findStudentwithSecondHighestScore:

This method will take an array of Student objects as an input parameter. This method will return the object of the second highest score student from the array of Student objects who are not from the dayScholar.

If no Student is a dayScholar in the array of Student objects, then the method should return null.

Note : All the searches should be case insensitive.

The combination of dayScholar and score for each student is always unique.

The above mentioned static methods should be called from the main method.

For findCountOfDayscholarStudents method - The main method should print the returned count as it is if the returned value is greater than 0, else it should print "There are no such dayscholar students".

For findStudentwithSecondHighestScore method - The main method should print the rollNo, name and score in the below format from the returned object if the returned value is not null.

rollNo#name#score

If the returned value is null, then it should print "There are no student from non day scholar"

Before calling these static methods in main, use Scanner object to read the values of four Student

objects referring attributes in the above mentioned attribute sequence.

Consider below sample input and output:

Input:

```
1001
Ashwa
IT
85
true
1002
Preeti
IT
70
false
1003
Uma
ECE
85
false
1004
Akash
EEE
90
true
```

Output:

```
2
1002#Preeti#70.0
```

Question 97

Create a class called Employee with the below attributes:

```
employeeId - int
name - String
branch - String
rating - double
company Transport - boolean
```

The above attributes should be private. write getters, setters and parameterized constructor as required.

Create class MyClass with main method.

Implement two static methods findCountOfEmployeesUsingCompTransport and findEmployeeWithSecondHighestRating in MyClass class.

findCountOfEmployeesUsingCompTransport method:

This method will take an array of Employee object and a String parameter as input parameters.

This method will calculate and return the count of Employees who are all using company transport in the given branch (String parameter passed).

If no Employee in the given branch using company transport in the array of Employee

objects, then

the method should return 0.

findEmployeeWithSecondHighestRating method:

This method will take an array of Employee objects as an input parameter. This method will return

the object of the second highest rating employee from the array of Employee objects who are not using company transport.

If all Employees using company transport in the array of Employee objects, then the method should return null.

Note: All the searches should be case sensitive, companyTransport and rating combination of each Employee is unique.

For findCountOfEmployeesUsingCompTransport method, The main method should print the returned count as it

is if the returned value is greater than 0, else it should print "No such Employees".

Ex: 2 , where 2 is the count

For findEmployeeWithSecondHighestRating method: The main method should print the employeeId and name from

the returned object if the returned value is not null. If the returned value is null, then it should

print "All Employees using company transport".

Ex: 1003

Uma

where 1003 is the employeeId and Uma is the name.

Before calling these static methods in main, use Scanner object to read the values of four Employee

objects referring attributes in the above mentioned attributes sequence. Next, read the value of String

parameter for capturing branch.

Input :

1001

Ashwa

IT

5

true

1002

Preeti

IT

4

true

1003

Uma

Admin

3

false

1004

Akash

Hardware

4.5

false

IT

Output :

2
1003
Uma

Question 98

Create a class Employee with the below attributes:

EmpId - int
EmpName - String
Dept - String
Rating - int
Salary - int

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create class Solution with the main method.

Implement two static methods - findDept and findSecondHighestSalary in Solution class.

findDept method:

This method will take two input parameters - array of Employee objects and int parameter. The method

will return the dept array attribute from employee objects whose salary is matched with int parameter and rating greater than equal to 3. If no data found of beach objects, then the method should return null.

findSecondHighestSalary method:

This method will take two input parameters - array of Employee objects and string parameter. The method

will return the second highest salary from employee objects whose dept is matched with string parameter.

If no data found of beach objects, then the method should return 0.

Note: No two employees have same rating. All the searches should be case insensitive.

These above mentioned static methods should be called from the main method.

For findDept method - The main method should print the dept from the Employee array. If return null, then you should print

"No Department found".

For findSecondHighestSalary method - The main method should print the 2nd highest salary from the Employee array with

the following condition. If return 0, then you should print "No data found".

Input ->

101
Arijit
Computer
4
35000
102
Rakesh
Electronics
2
18000
103
Mahima

```

Mechanical
3
35000
104
Saniya
Mechanical
5
68000
105
Rajesh
Computer
1
30000
35000
Mechanical
Output ->
Computer, Mechanical
35000

```

Question 99

create a class Footwear which consists of the below attributes.

```

footwearId=int
footwearName=String
footwearType=String
price =int

```

the above attributes should be private.

write getter and setter and parametrised constructor as required.

create the class footwearProgrammm with the main method.

implement the 2 static methods.getCountByType and
getSecondHighestPriceByBrand in the Solution class.

getCountByType method:

this method will take two input parameters.

array of the Footwear objects and string parameter footwear type.

this method will return the count of the footwears from array of the
footwear objects for the given type of footwear.

if no footwear with the given footwear type is found in the
array of footwear abjects,then the method should return 0.

getSecondHighestPriceByBrand method:

this method will take 2 input parameters-array of footwear objects and string parameter
inputFootwearName.the method

will return the second highest footwear objects based on the price from the array of
the Footwear objects

whose brand name matches with the input string parameter.

if no footwear with the given brand is present in the array of the footwear objects,the
the method

should return null.

NOTE: no two footwear objects would have the same footwearId.All the searches should be
case insensitive.

the above mentioned static methods should be called from the main method.

for getCountByType method- the main method should print the count of the footwears ,if the returned value is greater than zero. or it should print "Footwear not available";
for getSecondHighestPriceByBrand method-The main method should print price from the returned footwear objects
if the returned footwear object is not null.else it should print "Brand not available".
for example.

112

ABC

25555

where 112 is the footwear id,ABC is brand name,25555 is price.

consider the sample input and output.

100

Sketchers

sneakers

12345

103

Puma

running shoes

10099

102

reebok

Running shoes

5667

101

Reebok

running shoes

5656

99

reebok

floaters

5666

Running shoes

reebok

Sample output:

3

99

reebok

5666

Sample input2:

100

Puma

sneakers

12345

101

Puma

sneakers

10099

102

Puma

sneakers

5000

```
102
Reebok
sneakers
8000
104
Puma
floaters
2000
running shoes
bata

Sample output:
Footwear not available
Brand not available
```

Question 100

Create a class Hotel with the below attributes:

```
hotelId - int
hotelName - String
dateOfBooking - String (in the format dd-mon-yyyy)
noOfRoomsBooked - int
wifiFacility - String
totalBill- double
```

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - noOfRoomsBookedInGivenMonth and searchHotelByWifiOption in Solution class.

noOfRoomsBookedInGivenMonth method:

This method will take two input parameter - array of Hotel objects and a String parameter.

The method will return the total numbers of rooms booked from array of Hotel objects for the given month(String parameter passed).

If no rooms are booked for the given month in the array of Hotel objects, then the method should return 0.

searchHotelByWifiOption method:

This method will take two input parameter - array of Hotel objects and a String parameter

The method will return Hotel object with second highest totalbill, from the array of Hotel objects where wifiFacility attribute

matches with the given wifi facility(String parameter passed).

If no Hotel with the given wifi option is present in the array of Hotel objects, then the method should return null.

Note :

No two Hotel object would have the same hotelId.

No two Hotel object would have the same totalbill.

The Array either has at least two objects with specified wifi option OR no object with specified wifi option.

All the searches should be case insensitive.

dateOfBooking is stored in the format dd-mon-yyyy(eg. 01-Jan-2022)

The above mentioned static methods should be called from the main method.

For noOfRoomsBookedInGivenMonth method - The main method should print the total number of booked rooms as it is,
if the returned value is greater than 0, else it should print "No rooms booked in the given month"

For searchHotelByWifiOption method - The main method should print the hotelId from the returned Hotel object if the returned value is not null.

If the returned value is null then it should print "No such option available".

Before calling these static methods in main, use Scanner object to read the values of four Hotel objects referring attributes in the above mentioned attribute sequence.

Next, read the value of two String parameters for capturing the month and wifi option .

Consider below sample input and output:

Input1:

101

Best Stay

01-jan-2022

10

Yes

20000

102

Apple Stay

12-Feb-2022

3

Yes

4000

103

Accord

11-May-2022

5

Yes

15000

104

Royal Park

22-Dec-2021

7

Yes

12000

May

Yes

Output1:

5

103

Input2:

101

Best Stay

01-jan-2022

10

Yes

20000

102

Apple Stay

12-Feb-2022

3

Yes

4000

103

Accord

11-May-2022

5

Yes

15000

104

Royal Park

22-Dec-2021

7

Yes

12000

May1

Yes1

Output2:

No rooms booked in the given month

No such option available

Sample code snippet for reference:

Please use below code to build your Solution.

import java.util.Scanner;
public class Solution
{
public static void main(String[] args)
{
//code to read values
//code to call required method
//code to display the result
}
//code the first method
//code the second method
}
//code the class

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like nextInt(), nextDouble() etc.

Usually, this is not an issue, but this may be visible while calling nextLine() immediately after those methods.

Consider below input values:

1001

Savings

Referring below code:

```
Scanner sc = new Scanner(System.in);  
int x = sc.nextInt();
```


String str = sc.nextLine(); -> here we expect str to have value Savings. Instead it may be "".

If above issue is observed, then it is suggested to add one more explicit call to nextLine() after reading numeric value.

Question 101

Create a class AutonomousCar with the below attributes:

carId - int
brand - String
noOfTestsConducted - int
noOfTestsPassed - int
environment - String

Write getters, setters for the above attributes . Create constructor which takes parameter in the above sequence except grade.

Create class Solution with main method. Implement two static methods - findTestPassedByEnv and updateCarGrade in Solution class.

findTestPassedByEnv method:

This method will take two input parameters -array of AutonomousCar objects and string parameter environment. The method will return the sum of the noOfTestsPassed attribute from autonomousCar objects for the environment passed as parameter.

If no autonomousCar with the given environment is present in the array of AutonomousCar objects, then the method should return 0.

updateCarGrade method:

This method will take a String parameter brand, along with the array of AutonomousCar objects. The method will return the autonomousCar object, if the input String parameter matches with the brand attribute of the autonomousCar object.

Before returning the object, the grade should be derived based on the rating calculation mentioned below. This grade value should be assigned to the object. If any of the above conditions are not met, then the method should return null.

The grade attribute should be calculated as follows: rating .(noOfTestsPassed * 100)/noOfTestsConducted

If the rating > = 80 then grade should be 'A1', otherwise the grade should be 'B2'.

The above mentioned static methods should be called from the main method. For findTestPassedByEnv method - The main

method should print the totalTestPassed as it is, if the returned value is greater than 0, or it should print "There are no tests passed in this particular environment".

For updateCarGrade method - The main method should print the brand and grade of the returned autonomousCar object.

The brand and grade should be concatenated with :: while printing. eg:- Tesla::A1, where Tesla is the brand and A1 is the grade.

If the returned value is null then it should print "No Car is available with the specified brand". Before calling these static methods in main, use Scanner object to read the values of four autonomousCar objects referring attributes in the above mentioned attribute sequence (except grade attribute). Next, read the value for environment and brand.

Input

100

Tesla

1000

500

Hills

200

Ford

2000

1500

Desert

300

Royce

3000

1700

Hills

400

Mercedes

1000

400

Desert

Desert

Mercedes

Output

1900

Mercedes::B2

Question 102

Create a class NavalVessel with the below attributes:

vesselId - int

vesselName - String

noOfVoyagesPlanned - int

noOfVoyagesCompleted - int

purpose - String

Write getters, setters for the above attributes. Create constructor which takes parameter in the above sequence except classification.

Create class Solution with main method. Implement two static methods - findAvgVoyagesByPct and findVesselByGrade in Solution class.

findAvgVoyagesByPct method:

This method will take two input parameters -array of NavalVessel objects and int parameter percentage. The method will return the average of the noOfVoyagesCompleted attribute from NavalVessel objects which meets the percentage requirement as mentioned below. The percentage for each navalVessel is calculated as $\text{percentage} = (\text{noOfVoyagesCompleted} * 100) / \text{noOfVoyagesPlanned}$. If the above calculated percentage is greater than or equal to the percentage passed as the parameter, then those objects should participate in the average calculation. If there are no objects with the mentioned criteria, then the method should return 0.

findVesselByGrade method:

This method will take a String parameter purpose, along with the array of NavalVessel objects. The method will return the NavalVessel object, if the input String parameter matches with the purpose attribute of the NavalVessel object. Before returning the object, the classification should be derived as mentioned below. This classification value should be assigned to the object before returning. If any of the above conditions are not met, then the method should return null. The classification attribute should be calculated based on the percentage calculation mentioned above.

If the percentage = 100 then classification should be 'Star', if percentage is between 99 and 80 then classification should be 'Leader', if percentage is between 79 and 55 then classification should be 'Inspirer' otherwise the classification should be 'Striver'. If the returned value is null then it should print "No Naval Vessel is available with the specified purpose".

Before calling these static methods in main, use Scanner object to read the values of four NavalVessel objects referring attributes in the above mentioned attribute sequence (except the classification attribute). Next, read the value for percentage and purpose.

Input

111

Rojer

100

100

Passenger

222

Kalam

200

100

Goods

333

Yashwin

400

300

Army

444

Thanishwini

500

500

Welfare
75
Army
Output

300
Yashwin%Inspirer

Question 103

Create a class Institution with below attributes:

```
institutionId - int
institutionName - String
noOfStudentsPlaced - int
noOfStudentsCleared- int
location - String
grade - String
```

Write getters, setters for the above attributes.

Create constructor which takes parameter in the above sequence except grade.

Create class Solution with main method.

Implement two static methods - FindNumClearancedByLoc and UpdateInstitutionGrade in Solution class.

FindNumClearancedByLoc method:

```
-----
This method will take two input parameters - array of Institution objects and string
parameter location.
The method will return the sum of the noOfStudentsCleared attribute from institution
objects for the location passed as
parameter. If no institution with the given location is present in the array of
institution objects, then the method
should return 0.
```

UpdateInstitutionGrade method:

```
-----
This method will take a String parameter institutionName, along with the array of
Institution objects.
The method will return the institution object, if the input String parameter matches
with the institutionName attribute of the
institution object. Before returning the object, the grade should be arrived based on
the rating calculation mentioned below.
This grade value should be assigned to the object.If any of the above conditions are
not met, then the method should return
null.
```

* The grade attribute should be calculated as follows:

Rating=(noOfStudentsPlaced * 100)/noOfStudentsCleared

If the rating >= 80 , then grade should be 'A'.

Else, then grade should be 'B'

****Note:****

```
-----
No institution object would have the same value for institutionName attribute.
All institution object would have the noOfStudentsPlaced value lesser than
noOfStudentsCleared value.
```

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For FindNumClearancedByLoc method - The main method should print the noOfClearance as it is, if the returned value is greater than 0, or it should print "There are no cleared students in this particular location".

For UpdateInstitutionGrade method - The main method should print the institutionName and grade of the returned Institution object. The institutionName and grade should be concatenated with :: while printing. eg:- TCS::A, where TCS is the institution name and A is the grade.

If the returned value is null then it should print "No Institute is available with the specified name".

Before calling these static methods in main, use Scanner object to read the values of four Phone objects referring attributes in the above mentioned attribute sequence (except grade attribute). Next, read the value for location and institutionName.

Input

```
-----
111
Amrita
5000
10000
Chennai
222
Karunya
16000
20000
Coimbatore
333
AppleTech
10000
12000
Chennai
444
Aruna
6000
10000
Vellore
Chennai
Karunya
```

Output

```
-----
22000
Karunya::A
```

Starter Code

```
=====
import java.util.Scanner
public class Solution
{
    public static void main(String[] args)
    {
```

```
//code to read values
//code to call required method
//code to display the result
}
public static int findNumClearancedByLoc(Institution[] instArray, String location)
{
//method logic
}
public static Institution updateInstitutionGrade(Institution[] instArray, String
instName)
{
//method logic
}
}
class Institution
{
//code to build Institution class
}
```

Question 104

Create a class Student with below attributes:

```
rollNo - int
name - String
subject - String
grade - char
date - String [DD/MM/YYYY]
```

The above attributes should be private. Write Getter and Setter and parametrized constructor as required.

Create class Solution with main method.

```
*****
```

Implement one static method: findStudentByGradeAndMonth Solution Class.

findStudentByGradeAndMonth Method:

This method will take an array of Student objects, char value as grade and int value as month

for input parameters. The method will find out all Students from the given grade and month.

This method will return array of Student object ascending based on their rollNo if found.

If there is no Student that matches then the method should return null.

for this method- main method should print Student name, subject and total student found [The

length of the list], if the returned value is not null. If the returned value is null then

main method should print "No student found".

NOTE:

1. For Taking char as input use `sc.nextLine().charAt(0)`
2. To match/check the month You have to convert int month in the parameter to String.
3. No need to count the Student array if return not null just print the array length.

Consider the below input and output:

input1:

111

Arijit

Math

B

22/09/2023

101

Priyanka

English

A

30/03/2022

107

Shreosi

History

C

13/05/2022

105

Tatan

Physics

A

27/03/2022

A

3

output1:

Priyanka

English

Tatan

Physics

2

input2:

111

Sohel

Math

B

22/09/2022

101

Priyanka

English

A

30/03/2022

107

Gopa

History

C

12/05/2022

105

Kamal

Physics

A

27/03/2022

A

7

output 2:

No student found

Question 105

Create a class Sim with below attributes:

simId - int

customerName - String

balance - double

ratePerSecond - double

circle - String

Write getters, setters and parameterized constructor as required.

Public class Solution is already created with main method.

Code inside main method should not be altered else your solution might be scored as zero. You may copy the code from main method in eclipse to verify your implementation.

Implement static method - transferCustomerCircle in Solution class.

This method will take first parameter as array of Sim class objects, second parameter as circle to be transferred (which is String parameter circle1) and third parameter as new circle (which is String parameter circle2).

Method will transfer the customer to new circle (circle2), where the circle attribute would match second parameter (circle1).

Method will return array of Sim objects for which circle is transferred.

Return array should be sorted in descending order of ratePerSecond (assuming ratePerSecond is not same for any of the Sim objects).

This method should be called from main method and display the simId, customerName, circle and ratePerSecond of returned objects (as per sample output).

Main method mentioned above already has Scanner code to read values, create objects and test above methods. Hence do not modify it.

Consider below sample input and output:

Input:

1


```
raj
100
1.5
KOL
2
chetan
200
1.6
AHD
3
asha
150
1.7
MUM
4
kiran
50
2.2
AHD
5
vijay
130
1.8
AHD
AHD
KOL
```

Output:

```
4 kiran KOL 2.2
5 vijay KOL 1.8
2 chetan KOL 1.6
```

Question 106

Create a class Bill with below attributes:

```
billNo- int
name - String
typeOfConnection - String
billAmount - double
status - boolean
```

where billNo is the bill number, name is the name of the customer, typeOfConnection is the type of the connection (prepaid, postpaid), billAmount is the bill amount and status is whether the bill is paid or not (if paid then value is TRUE else value is FALSE).

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - findBillWithMaxBillAmountBasedOnStatus and getCountWithTypeOfConnection in Solution class.

findBillWithMaxBillAmountBasedOnStatus method:

This method will take an array of Bill objects and a boolean parameter as parameters. The method will return bill object array in ascending order of their bill number from the array of Bill objects whose bill amount is maximum in the array with the status attribute that matches with the input parameter.

If no Bill with the given status is present in the array of Bill objects, then the method should return null.

getCountWithTypeOfConnection method:

This method will take two input parameters - array of Bill objects and string parameter (for type of connection).

The method will return the count of bills from array of bill objects for the given type of connection.

If no bill with the given type of connection is present in the array of bill objects, then the method should return 0.

Note :

Two bill object can have the same bill amount.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findBillWithMaxBillAmountBasedOnStatus method - The main method should print the billNo followed by # and name from the returned Bill object array if the returned value is not null.

If the returned value is null then it should print "There are no bill with the given status".

For getCountWithTypeOfConnection method - The main method should print the count of bills as it is, if the returned value is greater than 0, otherwise it should print "There are no bills with given type of connection".

Before calling these static methods in main, use Scanner to read the number of object and objects to read the values of Bill objects referring attributes in the above mentioned attribute sequence.

Next, read the value for status and typeOfConnection.

Consider below sample input and output:

Input:

```
4
111
Aman Mittal
Prepaid
914.25
true
222
Rekha Kumar
Prepaid
1425.75
false
333
Samyra Gupta
Prepaid
1305.00
```

```
true
444
Mohit Saxena
Postpaid
1300.50
false
false
Prepaid
```

Output:

```
222#Rekha Kumar
```

```
3
```

```
*****
```

```
-----
Sample code snippet for reference:
Please use below code to build your solution.
```

```
-----
import java.util.Scanner;
public class Solution
{
    public static void main(String[] args)
    {
        //code to read values
        //code to call required method
        //code to display the result
    }

    public static Bill[] findBillWithMaxBillAmountBasedOnStatus( /* required parameters to
    be added */ ) {
        //method logic
    }

    public static int getCountWithTypeOfConnection( /* required parameters to be added */ )
    {
        //method logic
    }
}

class Bill
{
    //code to build the class
}
-----
```

Note on using Scanner object:

Sometimes scanner does not read the new line character while invoking methods like nextInt(), nextDouble() etc.

Usually, this is not an issue, but this may be visible while calling nextLine() immediately after those methods.

Consider below input values:

```
1234
```

```
Merin Bakers
```

Referring below code:

```
Scanner sc = new Scanner(System.in);
int x = sc.nextInt();
```

String str = sc.nextLine(); -> here we expect str to have value Bakers name. Instead it may be "".

If above issue is observed, then it is suggested to add one more explicit call to nextLine() after reading numeric value.

Question 107

Create a class TravelAgencies with below attributes:

regNo - int
agencyName - String
packageType - String
price - int
flightFacility - boolean

Write getters, setters for the above attributes . Create constructor which takes parameter in the above sequence.

Create class Solution with main method. Implement two static methods - findAgencyWithHighestPackagePrice and agencyDetailsforGivenIdAndType in Solution class.

findAgencyWithHighestPackagePrice method:

This method will take array of TravelAgencies objects as an input parameter and return the highest package price from the given array of objects.

agencyDetailsForGivenIdAndType method:

This method will take three input parameters -array of TravelAgencies objects, int parameter regNo and String parameter packageType. The method will return the TravelAgencies object based on below conditions.

FlightFacility should be available.

The input parameters(regNo and packageType) should matched with the regNo and packageType of TravelAgencies object.

If any of the above conditions are not met, then the method should return null. Note : Same Travel agency can have more than one package type. Travel agency and package type combination is unique. All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For findAgencyWithHighestPackagePrice method - The main method should print the highestPackagePrice as it is.

For agencyDetailsForGivenIdAndType method -The main method should print the AgencyName and price of the returned object.The AgencyName and price should be concatenated with : while printing.

Input

123

A2Z Agency

Platinum

50000

true

345

```
SSS Agency
Gold
30000
false
987
Cox and Kings
Diamond
40000
true
888
Global Tours
Silver
20000
false
987
Diamond
-----
Output
-----
50000
Cox and Kings:40000
```

Question 108

Create a class Person with the following attributes:

```
firstName - String
lastName - String
age - int
gender - char ('M' for male and 'F' for female)
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Student that extends the Person class and adds the following attributes:

```
rollNo - String
course - String
semester - int
GPA - double
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Faculty that extends the Person class and adds the following attributes:

```
employeeId - String
department - String
designation - String
salary - double
```

Write getters, setters and parameterized constructor in the above mentioned attribute sequence as required.

Create a class Solution with the main method.

Implement two static methods - findHighestGPASStudent and findHighestPaidFaculty in Solution class.

findHighestGPASStudent method:

This method will take an array of Student objects as input.

The method will return the Student object with the highest GPA.

If the array is empty, the method should return null.

findHighestPaidFaculty method:

This method will take an array of Faculty objects as input.

The method will return the Faculty object with the highest salary.

If the array is empty, the method should return null.

These above mentioned static methods should be called from the main method.

For findHighestGPASStudent method - The main method should print the roll number, course and GPA of the returned Student object.

If the returned value is null then it should print "Sorry - No student is available".

For findHighestPaidFaculty method - The main method should print the employee ID, department and salary of the returned Faculty object.

If the returned value is null then it should print "Sorry - No faculty is available".

Before calling these static methods in main, use Scanner object to read the values of two Student objects and two Faculty objects referring attributes in the above mentioned attribute sequence.

Example Input:

First1

Last1

20

M

Roll1

Course1

2

3.8

First2

Last2

21

F

Roll2

Course2

3

3.9

Faculty1

Last3

35

M

ID1

Department1

Associate Professor

90000.0

Faculty2

Last4

40

F

ID2

Department2

Professor

100000.0

Output:

Roll2
Course2
3.9
ID2
Department2
100000.0

Question 109

Create a class Employee with below attributes:

employeeId - int
employeeName - String
age - int
gender - char
salary - double

where employeeId is the unique identifier of the employee, employeeName is the name of the employee, age is the age of the employee, gender is the gender of the employee and salary is the salary of the employee.

The above attributes should be private, write getters, setters and parameterized constructor as required.

Create class Solution with main method.

Implement two static methods - getEmployeeWithSecondLowestSalary and countEmployeesBasedOnAge in Solution class.

getEmployeeWithSecondLowestSalary method:

This method will take an array of Employee objects as a parameter.

The method will return the Employee object with the second lowest salary in the array of Employee objects.

If there are less than two employees in the array, then the method should return null.

countEmployeesBasedOnAge method:

This method will take two input parameters - array of Employee objects and an integer parameter (for age).

The method will return the count of employees from the array of Employee objects whose age matches with the input parameter.

If no employee with the given age is present in the array of Employee objects, then the method should return 0.

Note :

Two employee objects can have the same salary.

All the searches should be case insensitive.

The above mentioned static methods should be called from the main method.

For getEmployeeWithSecondLowestSalary method - The main method should print the employeeId followed by # and employeeName from the returned Employee object, if the returned value is not null.

If the returned value is null then it should print "Null".

For countEmployeesBasedOnAge method - The main method should print the count of employees as it is, if the returned value is greater than 0, otherwise it should print "No employee found for the given age".

Before calling these static methods in main, use Scanner to read the number of objects and objects to read the values of

Employee objects referring attributes in the above mentioned attribute sequence.

Consider below sample input and output:

Input:

```
4
101
John
30
M
10000.00
102
Samantha
25
F
15000.00
103
Alex
28
M
12000.00
104
Lisa
30
F
15000.00
30
```

Output:

```
103#Alex
2
```

Question 110

Create a class named "Employee" with the following attributes:

```
id (int)
name (String)
designation (String)
salary (double)
```

The above attributes should be private, and the class should have a parameterized constructor to initialize them.

Implement the following methods in the Employee class:

getSalary(): This method should return the salary of the employee.

getDesignation(): This method should return the designation of the employee.

Create a class named "Company" with the following attributes:

```
companyName (String)
employees (Employee array)
numEmployees (int)
```

The above attributes should be private, and the class should have a parameterized constructor to initialize them.

Implement the following methods in the Company class:

getAverageSalary(): This method should return the average salary of all the employees in the company.

getMaxSalary(): This method should return the highest salary of all the employees in the company.

getEmployeesByDesignation(String designation): This method should return an array of all employees with the given designation.

In the main method, use a Scanner object to read the values of the company name, number of employees, and the employee details (id, name, designation, and salary) in a loop. Once all employee details have been read, create a Company object and pass the appropriate parameters. Finally, call the three methods mentioned above and print the output.

Note:

All searches should be case-insensitive.

You may assume that the input is valid and in the correct format.

Sample input:

```
-----
Enter company name: ABC Company
Enter number of employees: 3
Enter employee details:
Employee 1:
Enter id: 101
Enter name: John Smith
Enter designation: Manager
Enter salary: 5000
Employee 2:
Enter id: 102
Enter name: Jane Doe
Enter designation: Engineer
Enter salary: 4000
Employee 3:
Enter id: 103
Enter name: Bob Johnson
Enter designation: Engineer
Enter salary: 4500
```

Sample Output:

```
-----
Average salary: 4500.0
Max salary: 5000.0
Employees with designation: Engineer
ID: 102, Name: Jane Doe, Designation: Engineer, Salary: 4000.0
ID: 103, Name: Bob Johnson, Designation: Engineer, Salary: 4500.0
```

Question 111

Create a class called BankAccount with the following attributes:

accountNumber (String)

accountHolderName (String)

balance (double)

Write getters, setters, and a parameterized constructor for the class.

Create a static method `transferFunds` in a separate class called `BankUtils`. The method should take four parameters:

```
fromAccount (BankAccount)
toAccount (BankAccount)
amount (double)
transactionCode (String)
```

The method should transfer the specified amount from the `fromAccount` to the `toAccount`, and return a `Transaction` object containing the `transactionCode`, the amount transferred, and the timestamp of the transaction. If the `fromAccount` has insufficient balance, the method should throw an `InsufficientBalanceException` with an appropriate error message.

Create a class called `Transaction` with the following attributes:

```
transactionCode (String)
amount (double)
timestamp (String)
```

Write getters and a parameterized constructor for the class.

Create an `InsufficientBalanceException` class that extends the `Exception` class.

The exception should have a constructor that takes a message as a parameter.

In the main method, create two `BankAccount` objects, transfer some funds between them using the `transferFunds` method, and print the transaction details to the console. Handle any exceptions.

Sample Input:

Account 1:

```
Account Number: 12345
Account Holder Name: John
Balance: 5000.0
```

Account 2:

```
Account Number: 67890
Account Holder Name: Jane
Balance: 10000.0
```

Transfer Details:

```
Amount: 2000.0
Transaction Code: T0001
```

Sample Output:

Before Transfer:

```
Account 1: John - 12345 - 5000.0
Account 2: Jane - 67890 - 10000.0
```

After Transfer:

```
Account 1: John - 12345 - 3000.0
Account 2: Jane - 67890 - 12000.0
```

Transaction Details:

```
Transaction Code: T0001
Amount Transferred: 2000.0
Timestamp: 2023-04-01 10:30:00
```

Question 112

create the class `Course` with the below Attributes.

```
courseId- int
courseName- String
```

```
courseAdmin- String
quiz- int
handson -int
```

The above methods should be private ,write getter and setter and parametrized constructor as required.

create class courseProgram with main method.

implement two static methods-

findAvgOfQuizByAdmin method:this method will take array of Course objects and a String value as input parameters. This method will find out Average (as int) of Quiz questions for given Course Admin (String parametre passed) This method will return Average if found.if there is no course that matches then the method should return 0.

sortCourseByHandsOn method:

This method will take an Array of Course Objects and int value as input parameters.

This methods should return an Array of Course objects in an ascending order of their handson which are less than the given handson(int parameter passed) value. if there is no such course then the method should return null.

The above mentioned static methods should be called from main methods.

for findAvgOfQuizByAdmin method: The main method should print the average if the returned value is not 0. if the returned value is 0 then it should print "No Course found."

for sortCourseByHandsOn method:

the main method should print the name of the Course from the returned Course object Array if the returned value is not null.if the returned value is null then it should print "No Course found with mentioned attribute."

input1:

```
111
kubernetes
Nisha
40
10
321
cassandra
Roshini
30
15
457
Apache Spark
Nisha
30
12
987
site core
Tirth
50
```

20
Nisha
17
output1:
35
kubernetes
Apache Spark
cassandra
input2:
111
kubernetes
Nisha
40
10
321
cassandra
Roshini
30
15
457
Apache Spark
Nisha
30
12
987
site core
Tirth
50
20
Shubhamk
5
output 2:
No Course found
No Course found with mentioned attributes.
